# Corridor, Not Factory: Trade Reorientation and the Missing Investment Response in Kazakhstan, 2022–2025

Zhanbolat Kakishev*

August 2026

*Preliminary draft — comments welcome — please do not cite*

**Abstract**

A large, sector-specific rise in demand for a tradeable good is expected to induce domestic investment—import substitution, value-chain entry, new capacity. After February 2022 Kazakhstan received such a shock: with Russia's direct trade disrupted, part of it was rerouted through neighbouring economies. We measure the shock at the product level (HS6, 2018–2025), estimate what Kazakhstan retains, and ask whether it induced investment; to our knowledge this is the first host-economy incidence analysis of the post-2022 rerouting. In 29 product lines, exports to Russia rise sharply and discontinuously in 2022m5. The raw increase is roughly tenfold; a difference-in-differences against control lines, which nets out common shocks, leaves a roughly threefold increase. Imports of the same goods rise in step, though the inbound series is dominated by a large pre-existing flow from China. Kazakhstan re-exports the goods with no sign of domestic transformation. The retained trade-and-logistics margin, 6–14% by the national-accounts convention, implies domestic value added of roughly 5–11 cents per dollar rerouted, against about three-quarters for a dollar of domestic manufacturing. We find no investment response—though, with four post-2022 years, the data rule out a large response rather than a modest step-up, and the null is over-determined by contemporaneous shocks. Deal-making in manufacturing, transport and logistics runs

*Nazarbayev University, `zhanbolat.kakishev@nu.edu.kz`. *Conflict of interest.* The author is employed by the Astana International Financial Centre (AIFC), which operates the Astana International Exchange and is institutionally adjacent to Kazakhstan's state investment vehicles; and previously worked with a consulting team on a Kazakhstan trade-strategy engagement. Neither role is used in this paper: all analysis is based on public data and on commercial deal databases to which the author has academic access, and the paper takes a positive rather than a prescriptive stance on Kazakh investment policy. Replication materials (code, query specifications, public data) are at `scripts/R/kz_passthrough/` and `scripts/R/kz_valueadd/`.

at its pre-2022 rate; 2022 is the weakest year in the series; and no transaction over 2015–2025 falls in the product lines the reorientation flows through. We read the null through a three-gate decomposition—irreversibility, market access (a customs union makes trade a substitute for production), and capital-market institutions. Two within-country comparisons—captive state capital abstained too, and the same institutions added assembly capacity for a *durable* auto-demand shock but none for the transitory one—are suggestive that the irreversibility gate binds, though neither is a clean test. Kazakhstan operates as a corridor, not a factory: the trade statistics move by an order of magnitude while the domestic economy barely registers it.

**Keywords:** trade reorientation, intermediated trade, investment under uncertainty, irreversibility, industrialisation, value added, institutional voids, transit economies, Kazakhstan. **JEL:** F14, F15, E22, D25, O14, O24, O33, P33.

# 1 Introduction

When demand for a country's tradeable output rises sharply and persistently, the textbook expectation is that supply follows: firms expand, new entrants appear, and—if the country was importing the good—some of the demand is met domestically. This logic underlies a large part of development economics, from infant-industry and import-substitution arguments to modern work on global-value-chain entry and the investment effects of trade shocks. Its premise is that a demand signal, once clear enough and durable enough, is acted on by investment.

After February 2022 Kazakhstan received an unusually clean version of such a signal. The disruption of direct trade between Russia and its principal partners produced a visible redirection of trade through third countries. Chupilkin et al. [2026] document the pattern: a sharp fall in EU exports to Russia alongside a matching rise in EU exports to Armenia, Kazakhstan and the Kyrgyz Republic, with product-level and mirror-statistics evidence of onward movement to Russia; Chupilkin et al. [2025] quantify the substitution. For Kazakhstan the reorientation is concentrated in a well-defined set of technology-intensive products and is large: at the monthly level, exports to Russia in these lines rose roughly tenfold. The routes, the demand and the products are all observable in the trade data.

This paper asks what Kazakhstan did with that signal. We proceed in three steps.

*First, we measure the shock.* Using UN Comtrade at HS6, annual for 2018–2025 and monthly for 2019–2025, we identify from the data a "surge basket" of 29 product lines in which the outflow to Russia and the inbound flow (Western reporters plus China) both at least doubled from 2019–2021 to 2022–2024. The outbound aggregate breaks sharply in 2022m5 (monthly sup-$F$ of 561); the inbound flow breaks in the same month but more weakly, and not at all in the noisier annual series, where a large pre-existing China flow dominates. Armenia and the Kyrgyz Republic show the same 2022 break with no comparable domestic reform. A difference-in-differences against control lines puts the outbound increase at $\gamma = 2.44$ (about threefold in levels), with the same picture from a pre-specified priority-list version; a placebo on the largest civilian lines shows the opposite sign.

*Second, we measure what Kazakhstan retains.* On the incremental flow to Russia (about \$479m over 2022–2025), the unit-value evidence shows no sign of transformation. Taking the retained trade-and-logistics margin from the Kazakhstan national-accounts convention (6–14%) and propagating it through the input–output table (OECD ICIO), we estimate domestic value added of about **5–11 cents** per dollar rerouted, against roughly three-quarters for a dollar of domestic manufacturing output. Customs duty on the incremental flow is well under 1% of customs revenue, and the flow is 0.2–0.3% of GDP.

*Third, we ask whether the shock induced investment*—and find no response the data can detect. We assemble deal-level data on mergers, acquisitions, and private-equity and venture transactions in Kazakhstan for 2015–2025 from three commercial databases (Capital IQ, PitchBook, Preqin), with FactSet and Dealroom extracts specified in Appendix A for cross-validation in the final draft, and supplemented by the transaction record of Kazakhstan's state investment corporation. Deal-making in manufacturing of tradeable goods, transport and logistics, and distribution runs at essentially its pre-2022 rate, and 2022 is the weakest year in the series; with four post-2022 years the test rules out a large response, not a modest step-up. Reported deal value rises, but is concentrated in ownership transfers of existing or distressed assets—two of the largest are nationalisations. Across the whole 2015–2025 window there is *no* transaction in the specific product lines that dominate the reorientation.

We interpret the non-response with a framework in which the supply response is the product of three gates: *irreversibility* (does anyone build for a shock this transitory and policy-contingent, given the option to serve the demand through trade?), *market access* (is domestic transformation required to reach the final market?), and *institutions* (can the projects that clear the first two be financed?). For Kazakhstan the market-access gate is plainly open—a customs union with the destination makes trade a close substitute for production—which is close to sufficient for a null, and which also forgoes the technology transfer a rules-of-origin regime would have forced [de Souza et al., 2024]. Two within-country comparisons, which we read as suggestive rather than dispositive (Section 8 sets out why neither is a clean test), point to the irreversibility gate being adverse as well: captive state capital abstained too, and the same institutions added capacity for a *durable* auto-demand shock but not the transitory one. The institutional gate then explains a separate fact—why the durable adjacent opportunities the shock reveals go unfinanced by private capital.

**Contribution.** The paper makes three contributions. (i) Its primary contribution is empirical: to our knowledge it is the first host-economy incidence analysis of the post-2022 trade rerouting, extending the existing literature—which establishes that rerouting happened and its scale—to what the intermediary retains, and to 2025. It shows that a large, sector-specific tradeable-demand shock produced no detectable investment response, a fact that speaks to when trade shocks induce a domestic supply response and when they do not [Juhász, 2018, Bloom et al., 2016, Juhász et al., 2024, Gopinath and Neiman, 2014]. (ii) It organises the two standard readings of a missing supply response—investment under uncertainty and irreversibility [Dixit and Pindyck, 1994], and institutional voids in the capital market [Khanna and Palepu, 1997, Rajan and Zingales, 1998]—into a single decomposition, adding a customs-union market-access gate under which trade substitutes for production, and uses two within-country comparisons—suggestive, not dispositive—to argue the irreversibility gate is adverse here too. The decomposition is an organising device rather than an estimated model. (iii) Methodologically, it combines public trade data, an input–output table, and multi-source commercial deal data into a replicable measure of what a transit economy captures, cross-checking the deal counts across databases rather than relying on one.

The rest of the paper is organised as follows. Section 2 describes the setting; Section 3 sets out the framework; Section 4 the data, including the deal-database construction; Section 5 measures the reorientation; Section 6 what Kazakhstan retains; Section 7 the (non-)investment response; Section 8 asks which gate binds; Section 9 the moderators and external validity; Section 10 where investment would pay off; and Section 11 concludes.

## 2 Setting

Kazakhstan is a founding member of the Eurasian Economic Union, which removes the customs frontier between Kazakhstan and Russia: goods that clear customs on entry to Kazakhstan move onward to Russia without a further border, and a supply from Kazakhstan to Russia is treated as intra-union. Kazakhstan is also the land bridge between China and Russia and, via the Trans-Caspian corridor, between China and Europe, though as a doubly landlocked economy its overland logistics costs are high and its supply-chain reliability limited [Arvis et al., 2010]. These features make it a natural conduit for trade that can no longer move directly, and they mean that acting as a conduit requires no transformation of the goods.

Several things other than the reorientation changed in Kazakhstan around 2022, and each is a potential confound. The "New Kazakhstan" reform programme associated with the presidency of Kassym-Jomart Tokayev—an outward-oriented agenda that gathered pace in 2022 with a June constitutional referendum and a November snap election—would raise trade and investment generally and gradually; we show below that what we observe is instead concentrated in a narrow set of products and discontinuous in mid-2022, that a civilian control basket moves far less, and that Armenia and the Kyrgyz Republic show the same break with no comparable reform. Three further shocks bear specifically on the *investment* null and cannot be differenced away: the January 2022 unrest and state of emergency ("Qandy Qantar"), in the first month of the post-period; the secondary-sanctions and compliance exposure that the rerouting itself created, which deters exactly the foreign acquirers who might have answered the shock; and the 2022–2025 wave of nationalisations, which both crowds the state investment pipeline and signals ownership risk to private entrants. We return to these where they matter, in Sections 7 and 8; they make the investment null *over*-determined rather than fragile, but they also mean we cannot cleanly attribute it to the trade shock alone.

# 3 A framework: when a demand shock builds capacity

Consider a potential entrant deciding whether to build capacity to serve a newly enlarged demand for a tradeable good that the country currently imports. The good can be re-exported to the final market without domestic transformation—this is the relevant case for a member of a customs union with the destination. The entrant has two ways to serve the demand. It can *trade*: buy the good abroad and resell it into the final market, earning a margin $\mu_T$ per unit, with no sunk cost and the ability to stop costlessly if demand falls. Or it can *produce*: sink an irreversible cost $I$ to build capacity, then earn a margin $\mu_P$ per unit on domestic output for as long as it operates, with the capacity worth only scrap if the demand disappears.

Let the enlarged demand support quantity $Q$ per period and persist each period with probability $\rho$ (so expected duration $1/(1-\rho)$), and let $r$ be the discount rate. Trading is always weakly profitable while the demand lasts, with present value $V_T = \dfrac{\mu_T Q}{1-\rho/(1+r)}$. Building is profitable only if

$$\underbrace{\frac{\mu_P Q}{1-\rho/(1+r)}}_{\text{PV of production margin}} - I > \underbrace{\Omega(\sigma,\rho)}_{\text{option value of waiting}} + V_T, \tag{1}$$

where $\Omega$ is the value of the option to defer, increasing in the uncertainty $\sigma$ around the shock's size and duration and decreasing in $\rho$ [Dixit and Pindyck, 1994]. Four observations follow.

*(i) The irreversibility gate.* The right-hand side of (1) is large exactly when the shock is transitory (low $\rho$), uncertain (high $\sigma$, e.g. because it is policy-contingent), and the technology is sunk-cost-intensive (high $I$). When any of these is extreme, no entrant builds—$R \approx 0$—regardless of the other conditions. This is a statement about *whether anyone builds*, and it does not depend on who the entrant is or how deep the capital market is.

*(ii) Trade is the outside option, not just "wait".* Because the good is re-exportable, the alternative to building is not idleness but $V_T$: the entrant can serve the demand through trade indefinitely. The demand shock raises the return to *both* margins. When the shock is transitory and $\mu_T$ is available at low cost, (1) fails and the supply response is trade rather than production. A corollary: if $\mu_P \leq \mu_T$—if Kazakhstan cannot even charge more for a

transformed good than it pays for the input—then building is dominated before $\Omega$ is added. The unit-value evidence in Section 6 is a direct measurement of this parameter.

*(iii) The market-access gate.* If serving the final market instead *required* domestic transformation—binding rules of origin, local-content rules, or simply a customs border that a pure re-export cannot cross—then $V_T \to 0$ and (1) is far more likely to hold. Membership of a customs union with the destination opens this gate and is what makes trade a viable substitute for production. The same logic runs through the imported-input intensity of the would-be domestic producer: where domestic production would itself lean heavily on imported components—the roundabout structure of Gopinath and Neiman [2014]—the value a local plant adds over a pure re-export is smaller still, and the transformation margin $\mu_P - \mu_T$ narrower. The gate also governs *dynamic* gains. de Souza et al. [2024] show, using Brazilian firm-to-firm technology-transfer records, that a foreign supplier facing a lower tariff shifts from *transferring* technology into the market toward *exporting* goods to it, and that learning from imported goods is less efficient than learning from a transfer—so import liberalisation raises static welfare but slows diffusion and productivity growth. An open market-access gate is the limiting case: with a pure re-export, the foreign firm has no reason to transfer anything, and the host forgoes not only the static value added of Section 6 but the learning that would come with a plant.

*(iv) The institutional gate.* Suppose (1) *does* hold for some project—a durable opportunity with modest sunk costs. It is still realised only if the entrant can finance $I$ and, for an outside equity investor, exit. Where the capital market lacks growth equity and a functioning exit, projects that clear the irreversibility gate are not funded; the capital that does move is captive—retained earnings, or a state or development-finance investor—and is allocated by criteria other than $R$ [Khanna and Palepu, 1997, Rajan and Zingales, 1998].

The three gates combine multiplicatively, so the investment response to the shock is

$$\begin{aligned} R \;=\; & \underbrace{f\big(\rho,\ \sigma^{-1},\ I^{-1}\big)}_{\text{irreversibility}} \times \underbrace{h(\text{transformation required})}_{\text{market access}} \\ & \times\ \underbrace{g(\text{financial depth, exit, private/state mix})}_{\text{institutions}}. \end{aligned} \tag{2}$$

The structure is multiplicative: a closed gate anywhere drives $R$ to zero, and the gates bear on different questions—the first two on whether production is ever the right way to serve the demand, the third on whether the projects that pass get built. Sections 5–8 show that for Kazakhstan the irreversibility and market-access gates are both adverse, so $R \approx 0$; Section 9 and Section 10 show that the institutional gate explains a separate fact—the durable adjacent opportunities the shock reveals go unfinanced.

**A mediation restatement.** The same structure can be put in mediation terms; we do so as an interpretive device, not as a model we estimate—the data below have no firm-level panel linking the trade response to the investment response. Let the demand shock $X$ bear on the investment response $Y$ through a direct path $X \to Y$ and an indirect path $X \to M \to Y$, where the mediator $M$ is the volume of re-export activity the shock generates. The shock raises $M$; higher $M$ substitutes for domestic capacity, so it works against $Y$. When the market-access gate is open ($h = 1$, no transformation required), trade is a close substitute and nearly all of the shock is carried by $M$: the channel that would raise $Y$ is barely opened, and the net effect on investment is close to nil. When the gate is closed ($h < 1$), $M$ absorbs less of the shock and part of $X$ bears on $Y$ directly, with that residual governed by the irreversibility parameters ($\rho$, $\sigma$, $I$)—whether the un-absorbed demand is large and durable enough to clear (1). Kazakhstan is the polar case: $h = 1$, and the unit-value wedge of Section 6 is a rough estimate of the margin on the traded path, which is no higher than the cost of the input—so little is left for the direct path even before the irreversibility gate is applied.

# 4 Data

**Trade.** UN Comtrade, HS 6-digit. Annual 2018–2025 for magnitudes; monthly 2019m1–2025m12 (authenticated API) for break timing and the event study. Kazakhstan stopped reporting monthly to Comtrade after February 2024, so the monthly series uses 2019–2023 and 2025; 2024 is covered by the annual data and by mirror flows. We use Kazakhstan's reported exports to Russia (complete) and, for the inbound side, a partner-reported (mirror) measure: the sum of exports to Kazakhstan reported by the EU-27, the United Kingdom, the United States, Japan, Korea, Switzerland, Norway ("Western") *and China*, because Kazakhstan's own import reports are incomplete for 2020–2022. China supplies about two-thirds of this inbound value and its flow to Kazakhstan was large and rising well before 2022; we therefore also track the Western component on its own, since that is the part that moves with the reorientation. Mirror data overstate Kazakh absorption for goods declared for Kazakhstan and then moved onward—the onward-movement bias that Chupilkin et al. [2026] use to identify the reorientation in the first place—a bias we return to in Sections 5.3 and 6. Neighbour comparisons (Armenia, the Kyrgyz Republic) and the economy-wide imports-by-chapter series used in Section 10 are from the same source.

**Product sets.** We take 75 HS6 lines: the 50 codes on the February-2024 EU/US/UK/JP *List of Common High Priority Items* (technology-intensive, potentially dual-use) plus 25 clearly civilian control codes. The **surge basket** is defined from the data as the 29 lines in which the mean inbound flow (West + China) and the mean outflow to Russia each at least doubled from 2019–2021 to 2022–2024, and exceeded \$0.2m (inbound) and \$0.1m (outbound) in the post period; the ratio is computed with \$10,000 added to numerator and denominator so that a line rising from near-zero does not qualify on a rounding artifact. Twenty-four of the 29 lines are on the priority list. Because the basket is selected on the post-period outcome, we treat the priority list as a pre-specified alternative throughout and benchmark the difference-in-differences against a selection-rule-matched permutation test (Section 5.3).

**Input–output.** OECD Inter-Country Input–Output tables, 2023 edition, Kazakhstan block, 45 ISIC rev. 4 industries, 2019. We form the domestic technical-coefficient matrix $A_d$ from the Kazakhstan-on-Kazakhstan intermediate block, the domestic Leontief inverse $L_d = (I - A_d)^{-1}$, and the direct domestic value-added coefficients $v_d$; the domestic value-added multiplier for sector $j$ is $(v_d' L_d)_j$. As a robustness check we repeat the calculation on the Kazakhstan Bureau of National Statistics symmetric input–output table (up to 68 products, 2023 reference year).

**Deal data.** Deal-level records of mergers, acquisitions, private-equity buyouts and growth transactions, and venture rounds involving Kazakhstan-domiciled targets, 2015–2025. The core extract combines Capital IQ, PitchBook and Preqin ($n \approx 500$ after de-duplication), classified to sectors and flagged for greenfield versus ownership transfer. For the final draft we reconstruct the same universe from FactSet and Dealroom as a cross-validation step; the exact query specification for all five databases (geography filter, deal types, date range, industry taxonomy, extraction date) is in Appendix A, and Table 3 reports the coverage of the three analysed here. All are commercial but widely available under academic licence, so the extract is replicable. The state investment corporation's project register—project, sector, region, financing year and cost—is a public disclosure of the fund (QIC/Baiterek); the replication package pins the release date of the register used here and lists the fields drawn from it.

**Macro.** World Bank World Development Indicators.

Table 1: Surge-basket trade, USD million per year. Inbound is a partner-reported (mirror) measure. China's 2024 figure is large and preliminary; its 2025 figure is not yet reported, so the 2025 West + China total is the Western component only.

| | 2018 | 2019 | 2020 | 2021 | **2022** | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|---|
| West + China → Kazakhstan (mirror) | 394 | 427 | 483 | 470 | **837** | 1,373 | 2,360 | 363 |
| *of which Western* | 162 | 162 | 168 | 184 | **337** | 444 | 419 | 363 |
| Kazakhstan → Russia | 17 | 12 | 7 | 8 | **128** | 145 | 119 | 133 |

# 5 The demand shock

## 5.1 Magnitudes

Table 1 reports the surge basket. Kazakhstan's exports to Russia in these lines were \$7–17m per year in 2018–2021; they were \$128m in 2022 and \$119–145m in 2023–2025—a rise of roughly an order of magnitude, and the sharpest, cleanest part of the picture. The inbound flow (West + China) ran near \$440m per year before 2022 and roughly \$0.8–2.4bn after, but that series is dominated by China, whose exports to Kazakhstan were already large and rising before the war (and are only partly reported for 2024–2025). The *Western* component of the inbound flow—the part aligned with the reorientation—rose from about \$170m per year to \$340–440m, a factor of about 2.3.

## 5.2 Timing and the event study

Monthly, the outbound series (Kazakhstan's exports to Russia) breaks sharply in 2022m5 (Bai–Perron, 95% CI 2022m4–2022m6; sup-$F$ of 561 against critical values in the low tens), following the March-2022 sanctions with a short lag and preceding every political-calendar date. The inbound series (West + China) breaks in the same month (95% CI 2022m4–2022m6; sup-$F$ of 24), with a second, smaller break in late 2023 as enforcement tightened; the Western component alone breaks in 2022m5 much more sharply (sup-$F$ of 241). On the *annual* panel the outbound and Western-inbound breaks are large (sup-$F$ of 151 and 142), but the West + China inbound break is not significant (sup-$F$ of 4), because the large pre-existing China flow swamps the 2022 movement at annual frequency—one more reason the reorientation is best seen in the outbound series and the Western inbound. The sup-$F$ statistics come from a mean-shift test on a persistent series and are evidence of a break, not a cardinal measure of its size. Figure 1 plots the monthly series; Figure 2 is a difference-in-differences event study at the HS6×year level (surge basket vs. control lines, reference year 2021, 95% confidence intervals clustered by HS6). The pre-treatment coefficients are

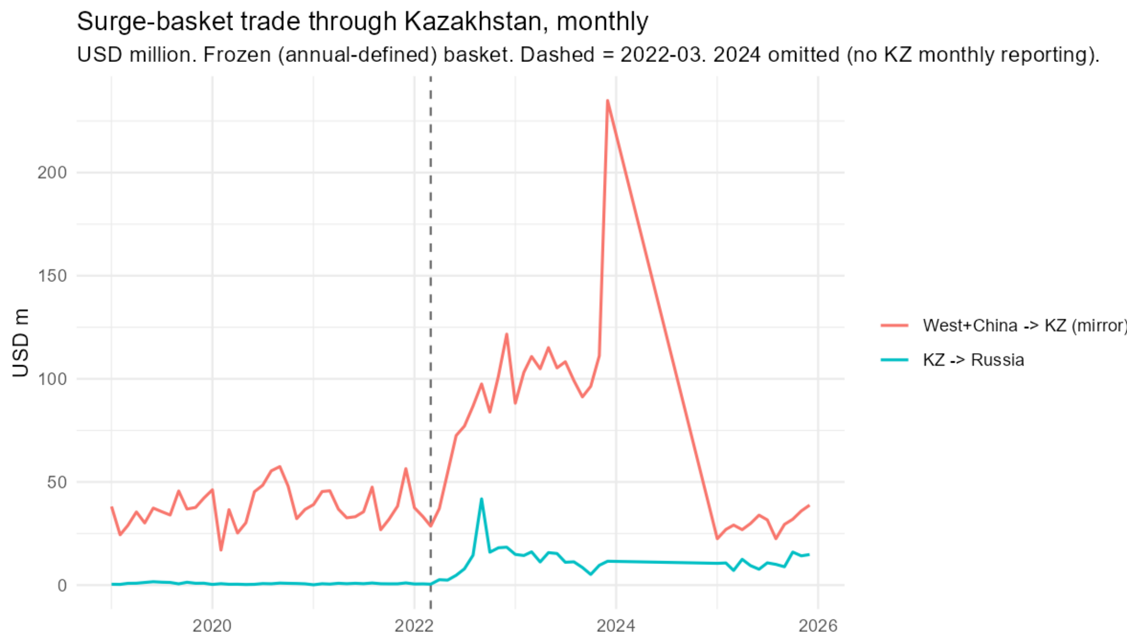


Figure 1: Surge-basket trade through Kazakhstan, monthly. USD million; blue = West + China mirrored exports to Kazakhstan, orange = Kazakhstan's exports to Russia. 2024 omitted (no Kazakhstan monthly reporting to Comtrade). Dashed line: March 2022. Source: UN Comtrade.

close to zero and jointly insignificant ($p = 0.78$ outbound, $p = 0.42$ inbound); the outbound coefficient jumps in 2022 and stays elevated through 2025, reaching about +2.4 asinh points (roughly a tenfold rise in level); the inbound coefficient follows a similar path with wider intervals. That the break precedes the June referendum, that the response is confined to a narrow product set, and that Armenia and the Kyrgyz Republic show the same 2022 break with no comparable reform (Section 5.3) are what separate the reorientation from the "New Kazakhstan" agenda.

## 5.3 Difference-in-differences

**Specification and identifying assumptions.** On the annual panel we estimate $\text{asinh}(y_{it}) = \gamma\,(\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$ with HS6 and year fixed effects and standard errors clustered by HS6 (75 lines, 8 years; reference year 2021). Identification requires parallel paths absent the shock (a joint test of the three pre-period interaction coefficients does not reject: $p = 0.42$ inbound, $p = 0.78$ outbound), no anticipation (dropping 2022 entirely, or setting the treatment year to 2023, leaves $\gamma$ essentially unchanged; see below), and no contamination between the surge and control baskets. The surge basket is *selected on the post-period outcome*, which inflates $\gamma$ mechanically; we address this with a selection-rule-matched permutation test rather than a uniform-random benchmark. Because treatment turns on once

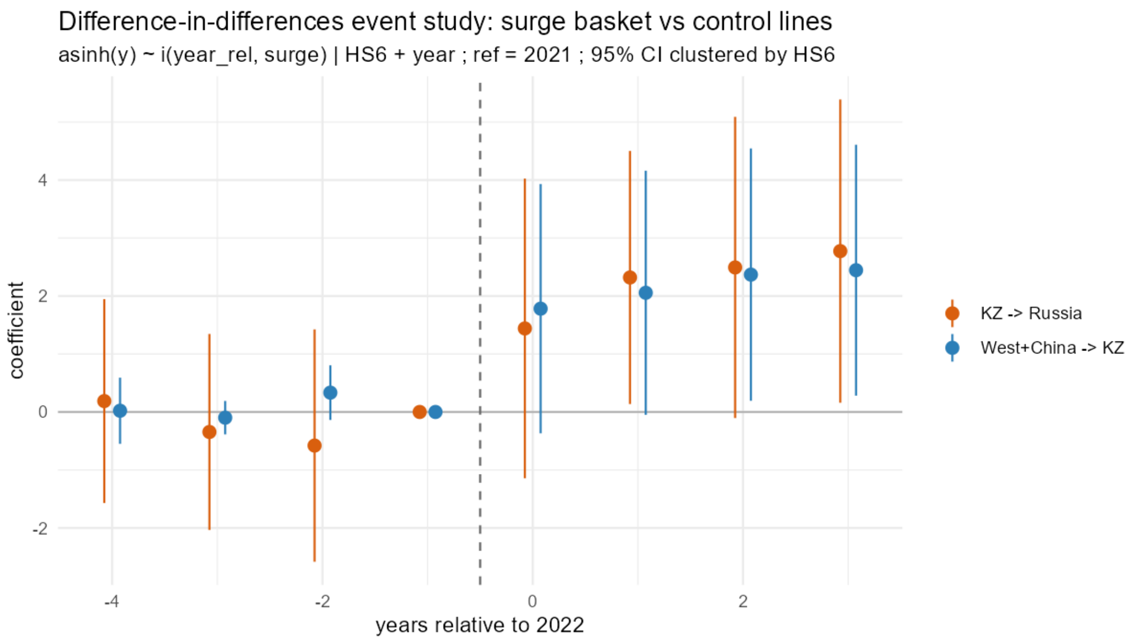


Figure 2: Difference-in-differences event study, surge basket vs. control lines: $\operatorname{asinh}(y_{it}) = \sum_{k\neq-1} \beta_k\,(\text{surge}_i \times \mathbf{1}[t-2022=k]) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, 95% confidence intervals clustered by HS6, reference year 2021. Source: UN Comtrade.

at a common date, we also report wild-cluster-bootstrap $p$-values, and we Holm-adjust across the four surge-basket outcomes.

**Results.** Table 2 reports $\gamma$ for the surge basket, the pre-specified priority list, and a placebo. On the surge basket, exports to Russia rise by $\gamma = 2.44$ ($p = 0.013$; wild-bootstrap $p = 0.010$; Holm-adjusted $p = 0.039$; Poisson in levels, about threefold, $p = 0.02$). The inbound flow (West + China) rises by $\gamma = 2.10$ ($p = 0.05$, not significant after Holm adjustment); the Western component alone by 1.73 ($p = 0.08$); Kazakhstan's own reported imports by 1.88 ($p = 0.004$, Holm $p = 0.016$). The pre-specified priority list, which is not selected on the outcome, gives the same pattern ($\gamma = 2.88$ outbound, 2.12 inbound). The outbound effect survives two demanding checks: adding size-decile$\times$year fixed effects (pre-2022 inbound size) leaves $\gamma = 2.29$ ($p = 0.003$), and dropping 2022 raises it to 2.71 ($p = 0.012$).

Three qualifications keep us from resting identification on this cross-sectional regression. First, the selection-rule-matched permutation: imposing the null by permuting the year labels within each HS6 and re-applying the doubling rule, the rule selects on average five lines (never as many as the 29 observed, so the *existence* of the surge basket is not a chance artifact, $p < 0.001$), but the placebo $\gamma$ on those lines averages 2.8, so the coefficient's *magnitude* is not separable from the selection. Second, the inbound criterion is close to redundant: 55 of the 75 lines clear the outbound threshold and only 41 the inbound one, so the operative screen is

Table 2: Difference-in-differences: $\gamma$ on treated$\times$post in $\text{asinh}(y_{it}) = \gamma(\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$, HS6 and year fixed effects, cluster-robust standard errors (by HS6) in parentheses. Surge-basket and priority-list rows have $N = 600$ (75 lines $\times$ 8 years); the placebo row has $N = 200$ (the 25 civilian lines). The surge basket is selected on the inbound (West + China) and outbound flows; the priority list (50 codes) is pre-specified.

| Treated set | Exports to Russia | Inbound (W+China) | Inbound (Western) | KZ-reported imports |
|---|---|---|---|---|
| Surge basket (29 HS6) | 2.44 (0.96)* | 2.10 (1.06) | 1.73 (0.98) | 1.88 (0.63)** |
| Priority list, pre-specified (50 HS6) | 2.88 (0.74)*** | 2.12 (0.71)** | 1.93 (0.67)** | 2.94 (0.53)*** |
| Placebo (largest civilian lines) | — | −0.95 (0.25)** | — | — |

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (cluster-robust; before multiple-testing adjustment). After Holm adjustment across the four surge-basket outcomes, exports to Russia and KZ-reported imports remain significant at 5%. Poisson in levels: surge basket 3.5× outbound ($p = 0.02$), 2.4× inbound ($p < 10^{-4}$).

the outflow to Russia. Third, the placebo row: assigning fake treatment to the largest civilian lines gives $\gamma = -0.95$ ($p = 0.001$) – the large consumer-goods lines contracted after 2022 as the tenge depreciation would predict – which shows the control pool is not homogeneous in trend. The surge coefficient's survival of the size-decile×year controls is what addresses this; we do not read the placebo as corroboration. Taken together, the cross-sectional difference-in-differences documents the outbound reorientation but does not identify it on its own; the structural breaks (Section 5) and the neighbour comparison below carry that weight. The mirror-gap outcome (partner-reported inflow minus Kazakh-reported imports) does not move ($\gamma = 0.54$, $p = 0.81$).

**The reform confound.** Three things separate the reorientation from the outward-oriented policy agenda of 2022. First, the civilian control basket tells the story a broad reform cannot: on the annual data its inbound flow shows no 2022 movement at all (sup-$F \approx 0.3$), and while its exports to Russia do shift—a general Russia-trade effect—the shift is a small fraction of the surge basket's (annual sup-$F$ of 34 versus 151; monthly, the surge-basket outbound break has sup-$F$ of 561), and the cross-sectional placebo runs the wrong way. Second, the outbound break is dated 2022m5, before the June referendum and the November election. Third, and most directly: Armenia's surge-basket exports to Russia rose from \$7–15m per year to \$71m in 2022 and \$93m in 2023, and the Kyrgyz Republic's from \$1–7m to \$12m and \$30m, on the same timing—neither country having a "New Kazakhstan" reform. The annual break statistics rest on eight observations and we read them as corroboration rather than inference; the direction is also asymmetric, in from sanctioning economies and out to Russia.

# 6 What Kazakhstan retains

## 6.1 The retained trade margin

We measure what Kazakhstan retains on the *incremental* flow—exports to Russia above the pre-2022 baseline, about \$479m cumulatively over 2022–2025—since that is the part the reorientation drove. Set against the incremental *Western* inbound of about \$887m the flow-through is roughly one half; set against the incremental West + China inbound of about \$3.2bn it is one sixth. The gap is domestic use, onward flow to other destinations, and measurement in the import series, which we cannot separate; the Western basis is the relevant one for the reorientation, and we use it here.

The unit-value evidence is consistent with a corridor and does not identify a retained margin on its own. On a c.i.f./f.o.b. basis—Kazakhstan's f.o.b. re-export price against the c.i.f. price it records paying on import, which includes freight into a landlocked economy—the median ratio across matched HS6×period cells is **0.73** (annual, $n = 112$; 0.70 monthly). On a like-for-like f.o.b. basis, against the exporter-reported price at the partner's border, the median is **1.6** (annual; 1.5 monthly), the difference being roughly the inbound freight. Neither ratio is distinctive—the civilian control basket gives 0.59 and 0.94 on the two bases—and the pass-through regression is weak on both (a slope of 0.06–0.48, significant only in the annual f.o.b. specification). Two features cut against domestic transformation: the re-exported tonnage is a small fraction of the imported tonnage in the median matched cell (weight ratio 0.08), so there is no weight gain; and the value-weighted aggregate wedge is below one, dominated by a handful of high-inbound-value cells and by outbound under-invoicing [Fisman and Wei, 2004]. What Figure 3 shows is wide dispersion with no mass at the high multiples a mark-up would produce. The wedge does not pin the retained margin, which we take from the national accounts next.

## 6.2 Input–output propagation

Let $m$ be the trade-and-freight margin Kazakhstan retains on a transiting good as a share of its value, $\bar{v}^{TT}$ the mean domestic value-added multiplier for its trade, transport and warehousing sectors, and $\bar{v}^{M}$ that for manufacturing. Domestic value added per dollar of gross rerouted flow is $m\,\bar{v}^{TT}$; per dollar of domestic manufacturing output it is $\bar{v}^{M}$ [Koopman et al., 2014, Johnson and Noguera, 2012]. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.79$ and $\bar{v}^{M} = 0.76$—close enough that the propagation step does little work: the result is essentially $m$ against the full domestic value $\bar{v}^{M}$.

We take $m$ from the Kazakhstan national-accounts convention for trade, insurance and

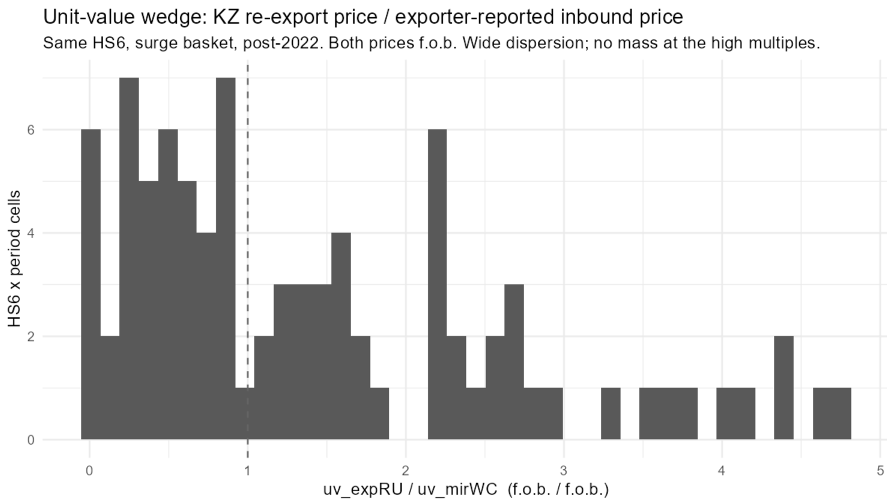


Figure 3: Unit-value wedge: Kazakhstan re-export price over the exporter-reported inbound price (f.o.b. basis), same HS6, surge basket, post-2022. Wide dispersion; no mass at the high multiples that domestic transformation would produce.

freight on goods trade, **6–14%**. (We do not use the matched-cell unit-value wedge to pin $m$: censoring negative-margin cells inflates the gross margin to about 34%, while the value-weighted aggregate is negative, and both are artifacts of the c.i.f./f.o.b. basis mismatch and of outbound under-invoicing rather than measures of domestic value retained.) Domestic value added per dollar rerouted is then $m\,\bar{v}^{TT} \approx$ **5–11%**, midpoint about **8%**, against **76%** per dollar of domestic manufacturing output—so a rerouted dollar creates on the order of a tenth as much domestic value as a produced one. Of the \$479m in incremental exports to Russia after 2022, Kazakhstan retains roughly **\$23–53m** as domestic value added.

The conclusion is not sensitive to the multipliers. Even at $\bar{v}^M = 0.40$ the ratio is about one to five. Recomputing $\bar{v}^{TT}$ and $\bar{v}^M$ from the Kazakhstan Bureau of National Statistics symmetric input–output table (68 products, 2023) rather than the OECD ICIO gives $\bar{v}^{TT} = 0.89$ and $\bar{v}^M = 0.74$; the rerouted-vs-produced ratio moves from about one-in-ten to about one-in-eight—still an order of magnitude. Nor does under-invoicing rescue it: for domestic value added to reach one-fifth of the gross flow the retained margin would have to be about 25%, nearly double the national-accounts convention, which would require exports to Russia to be under-recorded by roughly 40%—an order of magnitude larger than the mirror-gap discrepancies typical in this literature [Fisman and Wei, 2004].

## 6.3 Fiscal and macro

The reorientation-attributable flow—incremental exports to Russia over the pre-2022 baseline—is about \$479m cumulatively over 2022–2025 (\$107–134m per year). Customs duty on it, at an EAEU common external tariff of 4–8% on the share formally cleared in Kazakhstan, is on the order of **$22m over four years** (about $5m per year); net import VAT is close to zero because the onward supply to Russia is intra-union and zero-rated. Kazakhstan's annual customs revenue is of the order of $4–5bn, so the reorientation adds well under 1%. At the macro level the flow is 0.2–0.3% of GDP; the 2022 current-account surplus, the reserve accumulation of 2023–2025 and the rise in services value added are an order of magnitude too large to be driven by it.

## 7 The (non-)investment response

If the demand shock were seeding domestic value added, we would expect a rise in investment in the sectors the flow passes through. We find none—though, with only four post-2022 years, the deal-count test can rule out a large response, not a modest one.

Dealmaking in manufacturing of tradeable goods, transport and logistics, and wholesale distribution runs at **7.3 per year** in 2015–2021 and **7.5 per year** in 2022–2025 (a Poisson rate ratio of 1.03, 95% CI 0.63–1.65); 2022 itself is the weakest year in the series (Figure 4). Dropping the 2022 trough, 2023–2025 runs at 9.0 per year (rate ratio 1.24, 95% CI 0.74–2.01). The test has 80% power only against an increase of roughly 80% or more, so it excludes a surge in dealmaking but not a modest step-up. Three features of the data say more than the count. First, reported deal *value* in these sectors rises, but is concentrated in ownership transfers of existing or distressed assets—the ArcelorMittal Temirtau to Qarmet nationalisation (2023), Eurocopter Kazakhstan and the Lokomotiv rolling-stock plant (re-nationalisations, 2024), and a $0.8bn 2025 consolidation of rail-freight and forwarding assets—not new productive capacity. Second, greenfield and new-capacity transactions run at one to two per year throughout, with no post-2022 change. Third, across the whole 2015–2025 window the data contain **no** transaction in the specific product lines that dominate the reorientation—a zero that a modest-response CI does not soften.

The result holds across the three databases. Table 3 reports deal counts for the same Kazakhstan universe from each source; coverage differs in level—the transaction-oriented Capital IQ extract records more mid-market M&A, the venture-heavy PitchBook more small early-stage rounds, Preqin fewer of either—but none shows a post-2022 increase in manufacturing or logistics dealmaking. In the Capital IQ extract, which best captures the mid-market industrial deals that a genuine capacity response would generate, the count of manufacturing, transport and distribution deals is essentially unchanged before and after 2022 (5.1 versus 5.2 per year). The FactSet and Dealroom extracts specified in Appendix A are for cross-validation in the final draft.

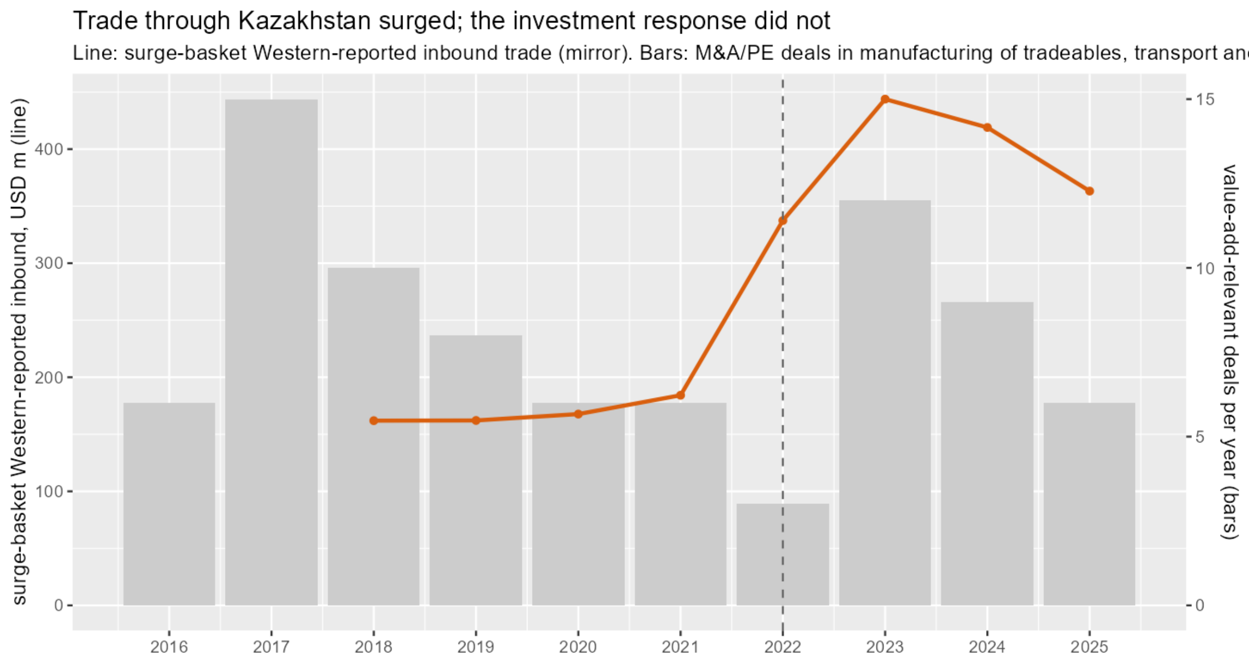


Figure 4: Trade through Kazakhstan surged; the investment response did not. Line: surge-basket *Western-reported* inbound trade (partner-reported mirror), USD million (left axis)—the West + China total is several times larger but was already growing before 2022. Bars: annual count of M&A/PE deals in manufacturing of tradeable goods, transport and logistics, and wholesale distribution (right axis). Dashed line: 2022. Source: UN Comtrade; Capital IQ/PitchBook/Preqin.

Kazakhstan is adding vehicle assembly (Chinese marques, KIA, Škoda) and some appliance capacity, but this predates and is largely independent of the reorientation, is kit assembly with low domestic value added, and serves the domestic and Eurasian consumer market rather than the flow of restricted components to Russia. The state investment corporation's post-2022 industrial pipeline is agri-processing and metallurgy, not corridor processing. The one investment signal that is reorientation-linked—the 2025 rail-freight consolidation, with Russia-linked asset names—deepens Kazakhstan's role as a forwarding operation, not a producing one.

# 8 Which gate binds?

The framework of Section 3 attributes the null to a closed gate. Which one? The market-access gate is plainly open: under the Eurasian Economic Union the good moves to Russia with no transformation and no border, and the unit-value evidence (Section 6) is consistent with no transformation occurring—so $V_T > 0$ and, with $\mu_P \lesssim \mu_T$, building is dominated in (1) before the option value is added. That is close to sufficient for $R \approx 0$ on its own. Two

Table 3: Deal counts for the Kazakhstan universe, by source and period

| Deals per year (unless noted) | Cap. IQ | PitchBk | Preqin | FactSet | Dealroom |
|---|---|---|---|---|---|
| All deals, 2015–21 | 24.0 | 14.0 | 3.1 | — | — |
| All deals, 2022–25 | 23.2 | 24.0 | 1.8 | — | — |
| Mfg + transport + distrib., 2015–21 | 5.1 | 1.9 | 0.3 | — | — |
| Mfg + transport + distrib., 2022–25 | 5.2 | 1.5 | 0.5 | — | — |
| Surge-basket lines, 2015–25 (total) | 0[a] | 0 | 0 | — | — |

[a] Two Capital IQ records carry an "electronic" industry tag—a steel-wire-rope manufacturer and a finance holding company—and are misclassifications; there are no transactions in the reorientation's product lines. FactSet and Dealroom columns pending (Appendix A).

within-country comparisons point to the irreversibility gate being independently closed as well. Neither is a clean identification—each holds some things fixed while others move—and we set out their limits.

*Captive capital abstained too.* Kazakhstan's state investment corporation (QIC/Baiterek) faces neither an external-finance constraint nor an exit constraint—the institutional gate does not bind for it in the way it binds for an outside equity investor. Yet its post-2022 deployment bypasses the reorientation almost entirely, and so does private capital (Table 4): of the state fund's 50 industrial and logistics projects financed in 2022–2025, one is in a surge-basket sector and two are in corridor logistics; the rest are domestic-market agri-processing, steel (the Qarmet rescue) and chemicals. This points away from a pure external-finance story: the investor least constrained by capital-market voids also did not build. Two caveats keep it from being decisive. QIC allocates under statutory sectoral programmes rather than scanning the opportunity set, so its silence in component manufacturing is partly a mandate artefact; and a state-owned vehicle is precisely the actor most exposed to secondary-sanctions and reputational risk from financing the rerouting of restricted goods, so its abstention is also consistent with a compliance constraint rather than an irreversibility one. The durable-shock comparison below is what separates these.

*A durable shock, the same institutions, a different outcome.* Over the same window, Kazakhstan's *durable* passenger-vehicle demand shock—Western marques withdrawing from Russia, a domestic consumption boom, and Eurasian market access—was met with real capacity: a combined Changan/Haval/Chery assembly facility of about 90,000 units, a KIA plant (about $200m), local assembly of several Škoda models, and a car-multimedia plant in Almaty (2024). The *transitory* component shock was met with no plant at all. We treat this as illustrative rather than as a controlled test, for three reasons. In the deal databases, auto-sector transactions actually *fell* after 2022 (six in 2015–2021, three in 2022–2025); the capacity evidence is from plant announcements and public reporting, not from the deal data, and no symmetric announcement search was run for the component sector. §7 notes that

Table 4: Investment 2022–2025 by relation to the reorientation

| | State fund (QIC/Baiterek) | | Private M&A / PE / VC | |
|---|---|---|---|---|
| Relation to the reorientation | projects | $m (fund) | deals | $m |
| Surge-basket sector | 1 | 4 | 1[a] | 10 |
| Corridor logistics / processing | 2 | 117 | 2 | 767 |
| Vehicles / machinery (domestic market) | 13 | 1,170 | 6 | 138 |
| Unrelated (agri, steel, chemicals, energy) | 34 | 453 | 192 | 10,373 |

*Notes.* The two dollar columns are on different bases and are not comparable across the divider: "$m (fund)" is QIC/Baiterek's own committed amount; "$m" is total disclosed transaction value, including co-investor and acquirer capital. Deal counts are the comparable quantities. [a] A steel-wire-rope manufacturer carrying an "electronic" industry tag; there is no genuine component-manufacturing transaction. Private-deal values are dominated by oil, mining and banking; the $767m in corridor logistics is a single 2025 rail-freight consolidation of existing assets.

much of the auto capacity predates the reorientation. And the two shocks differ not only in persistence $\rho$ but in sunk-cost intensity, in whether market access requires domestic content (EAEU local-content rules bind for vehicles, not for the re-export), and in the destination market. The comparison is suggestive that a durable, transformation-requiring shock draws capacity where a transitory re-export shock does not; it is not a clean $\rho$ experiment.

*The transitory condition is observed, not assumed.* The surge-basket flow to Russia peaked in September 2022 and then fell by more than a third over the second half of 2023 (from about $15m per month in 2023-H1 to about $9m in 2023-H2) as enforcement tightened (the EU's eleventh sanctions package, Council Regulation (EU) 2023/1214 of 23 June 2023, which introduced an anti-circumvention transit tool; sustained bilateral pressure on Kazakhstan through 2023). A firm observing this in real time saw the flow already receding within eighteen months—low $\rho$, and a $\sigma$ dominated by policy risk.

*The institutional gate explains the residual.* It does not explain the shock-specific null—captive capital abstained regardless—but it explains a distinct fact taken up in Section 10: the durable adjacent opportunities the shock reveals, for which the irreversibility gate is open (warehousing and transport support, with a domestic value-added multiplier of 0.82; machinery and fabricated-metal import substitution), are financed only slowly and only by directed state capital. The domestic private-equity market has no functioning exit—buy-back is the leading route—capital is dominated by state and development-finance money, and the 2022–25 pipeline is crowded by nationalisations. This is the institutional-voids configuration of Khanna and Palepu [1997] and Rajan and Zingales [1998], and it is the subject of related work.

Table 5: Moderators of the investment response, and Kazakhstan's values

| Moderator | Gate | Kazakhstan, 2022 | $R\uparrow$ when |
|---|---|---|---|
| Shock persistence $\rho$ (structural reallocation vs. sanctions-driven) | irreversibility | low (transitory) | high |
| Uncertainty $\sigma$ / policy & enforcement risk | irreversibility | high | low |
| Sunk-cost intensity $I$ of the relevant technology | irreversibility | high (component mfg) | low |
| Domestic transformation required to reach the market (rules of origin, local content, a border) | market access | none (customs union) | required |
| Financial depth / exit availability | institutions | shallow, no exit | deep |
| Private vs. state / directed capital mix | institutions | state-dominated | private-led |
| Existing productive base / absorptive capacity | institutions | near-zero (electronics output $\approx$ \$230m) | a base exists |

# 9 Moderators and external validity

Equation (2) makes the response a function of seven moderators, listed in Table 5 with the value each takes for Kazakhstan. The framework's testable content is that $R$ is positive only when the irreversibility and market-access gates are both open, and large only when the institutional gate is open as well; a country adverse on the first two should show a null irrespective of its institutions, and a country favourable on all three should show a supply response even with a shallow capital market.

*Within the reorientation group.* Armenia and the Kyrgyz Republic received the same shock—their surge-basket exports to Russia rose several-fold, breaking in 2022 on the same timing—and share Kazakhstan's value on every moderator: the shock is sanctions-driven, enforcement risk is high, and the customs union removes the need to transform. Two features are informative. First, both neighbours' flows are already receding by 2024–2025 (Armenia's surge-basket exports to Russia: \$71m, \$93m, \$42m, \$13m over 2022–2025; the Kyrgyz Republic's \$12m, \$30m, \$12m, \$12m), where Kazakhstan's persist (\$128m, \$145m, \$119m, \$133m). The neighbour reversion is consistent with the transitory reading of Section 8; the Kazakhstan asymmetry means the "same shock" claim holds for the 2022–2023 surge, not for its persistence. Second, we do not have deal-level data for Armenia or the Kyrgyz Republic—our commercial extracts are Kazakhstan-only—so we cannot test the investment null for them directly; what we can say is that neither has a domestic component-manufacturing sector to expand.

*Contrast cases.* Economies that drew capacity investment from a post-2020 tradeable demand shock did so with a favourable configuration on the first two gates, not necessarily the third. Vietnam and Mexico received China-plus-one and nearshoring reallocations widely read as *structural* [e.g. Juhász et al., 2024], and in autos the tightening of rules of origin under the United States–Mexico–Canada Agreement made domestic content a condition of market access (a closed market-access gate that forces $V_T \to 0$). Both invested in capacity despite financial markets that are not uniformly deep, which is the pattern the framework predicts: persistence and a transformation requirement do the work, financial development scales the response rather than gating it.

*What we cannot show here.* A clean separation of the irreversibility and institutional gates across countries would regress sector-level greenfield investment on shock persistence and financial depth in a panel of intermediaries and contrast cases. In the sample available to us the two are collinear—Kazakhstan, Armenia, the Kyrgyz Republic and the other Caucasus and Central Asian intermediaries are all financially shallow (private credit plus market capitalisation of 24–66% of GDP, all below the comparator median)—and country-level macro aggregates are too noisy to isolate a response of this size and in places run the other way: Armenia's gross fixed capital formation actually rose about three points of GDP after 2022. The within-country comparisons of Section 8 are what the argument leans on; a cross-country panel using greenfield-project data by sector is the natural extension.

# 10 Where investment would raise value capture

The reorientation is nonetheless informative for investment policy, because the trade data reveal which products have proven demand and proven routes through Kazakhstan. Table 6 sets three quantities side by side for the candidate sectors—the domestic value-added multiplier, the import-substitution gap (imports relative to domestic output), and the post-2022 growth in import demand—after removing the surge-basket lines from the electronics-adjacent chapters; we read them jointly rather than collapsing them into a single index (Figure 5).

Two groups stand out. The **logistics platform**—warehousing and transport support, and wholesale-trade infrastructure—has among the highest domestic value-added multipliers in the economy (0.82) and is the corridor's binding constraint, given the high overland logistics costs a landlocked transit economy faces [Arvis et al., 2010]; investment here creates domestic value and holds it whether or not the Russia flow persists. **Import substitution**

Table 6: Sector characteristics for value-adding investment, Kazakhstan. VA mult. = domestic value-added multiplier $(v'_d L_d)_j$ from the OECD ICIO block; Output = gross output, \$m; Imp./output = 2023 imports over domestic output; Imp. growth = ratio of 2023 to 2019 import value. Dashes: not applicable (platform sectors are not import-competing) or not computed. Surge-basket HS6 are netted out of the electronics and machinery chapters.

| | Sector | VA mult. | Output $m | Imp./ output | Imp. growth |
|---|---|---|---|---|---|
| Platform | Warehousing & transport support | 0.82 | 6,800 | — | — |
| | Wholesale & retail trade infrastr. | 0.82 | 81,000 | — | — |
| Substitution | Machinery & equipment n.e.c. | 0.80 | 1,160 | 7.9 | 1.27 |
| | Fabricated metal products | 0.81 | 1,850 | 1.8 | 1.22 |
| | Chemicals | 0.79 | 3,480 | 1.7 | 1.52 |
| Avoid | Computer, electronic & optical (surge basket) | 0.69 | 230 | — | — |
| | Motor vehicles & parts (kit assembly) | 0.69 | 1,530 | 6.1 | 2.55 |

in machinery, fabricated metal products and chemicals: Kazakhstan imports about eight times what it produces in machinery and roughly twice its output in the other two, the domestic value-added multiplier is around 0.80, and a nascent deal base exists. By contrast, localising the surge-basket products is the worst option—computer, electronic and optical manufacturing has the *lowest* value-added multiplier in the economy (0.69) and a near-zero domestic base—and the same is true of motor-vehicle kit assembly.

These opportunities are investable at scale only if the state investment corporation shifts from lead direct investor to co-investor behind private leads, with larger syndicated tickets and a credible exit—trade sale or listing—rather than the buy-back that currently dominates. That is a question about capital-market institutions, taken up in related work.

# 11 Discussion

Kazakhstan's trade with Russia and with the West in these products moved by an order of magnitude after 2022. The domestic economic content of that movement—value added about a tenth of the manufacturing equivalent, customs duty under one percent of revenue, a fraction of one percent of GDP, and no detectable investment response—is small. The country is functioning as a corridor, not a site of production.

The result speaks to a general question: when does a trade shock industrialise a country, and when does it merely pass through? The framework of Section 3 gives a three-part

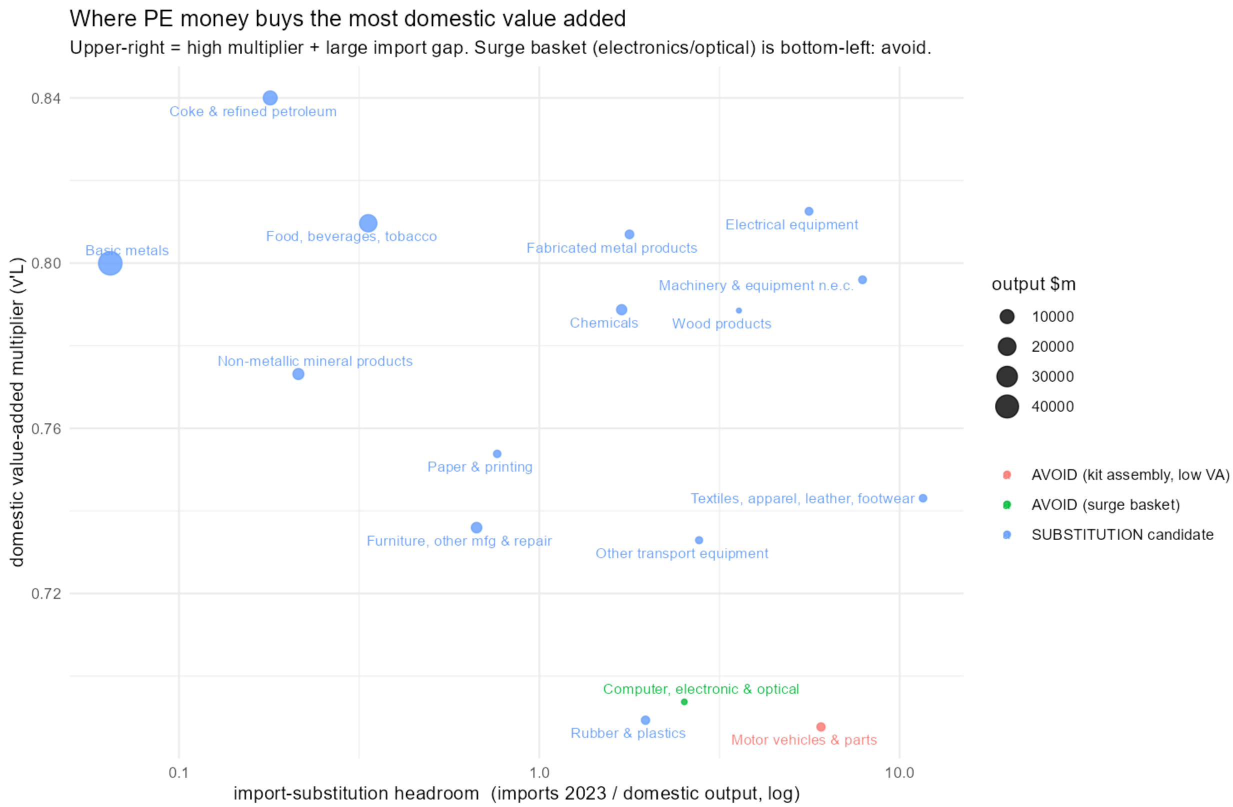


Figure 5: Where investment buys the most domestic value added: domestic value-added multiplier against import-substitution headroom (imports 2023 / domestic output, log scale), bubble size = domestic output. The surge-basket products (computer/electronic/optical) sit at the bottom.

answer. Production is a plausible response only when the demand cannot be met by trade alone (a closed market-access gate) and only when the shock is durable and certain enough, and the technology cheap enough to sink, that building beats waiting (an open irreversibility gate); and it is actually realised only where the capital market can fund the projects that pass (an open institutional gate). Kazakhstan is adverse on the first two: the customs union makes trade a close substitute for production, so the shock is largely absorbed into the trade channel, and the flow is transitory and policy-contingent, so even the un-absorbed demand does not clear the investment threshold. That the state investment corporation, which does not face the capital-market constraint, also did not build points to the block being upstream of finance, though its mandate and its own sanctions exposure cloud the comparison. The institutional gate then explains the missed alternative: the durable opportunities the shock reveals, in logistics and machinery, are left to slow, directed state capital. Armenia and the Kyrgyz Republic show the same 2022 surge and, on the evidence we have, no domestic component-manufacturing sector to expand; the framework's contrast cases—structural reallocations, or rules-of-origin regimes that force domestic content—are where a supply response does occur, even without a deep capital market. For the design of the sanctions that prompted the reorientation, the implication is that the intermediary retains little, so the economic pressure of enforcement falls on the final buyer and the original seller rather than on the transit country. And the static value Kazakhstan misses is only part of the cost: an open market-access gate also forgoes the technology transfer and learning that a rules-of-origin regime would have forced from foreign suppliers [de Souza et al., 2024], which is where the long-run cost of being a corridor sits.

**Limitations.** Two limitations are central. First, the argument for which gate binds rests on two within-country comparisons (captive versus constrained capital, and a durable versus a transitory shock) rather than on a cross-country panel, and neither is a clean experiment; the panel that would separate the irreversibility and institutional gates directly requires greenfield-project data by sector and country. Second, the investment null is *over*-determined—the January 2022 unrest, secondary-sanctions exposure and the nationalisation wave would each depress entry—so we bound its magnitude (Section 7) rather than attribute it to the trade shock alone.

Five measurement caveats bear on the estimates. The inbound side is partner-reported (mirror) data, which overstates Kazakh absorption for onward-moving goods. The inbound flow is dominated by a large pre-existing China component, so we measure value capture on the incremental Western flow and read the West + China inbound only at the product level, where line and year fixed effects absorb the China trend. The surge basket is selected on the post-2022 outcome; the pre-specified priority list gives the same picture, but the cross-sectional difference-in-differences should be read alongside the structural breaks and the neighbour comparison rather than on its own, and a placebo on the largest civilian lines is significant with the opposite sign, indicating the control pool is not homogeneous in trend. The retained margin is taken from the national-accounts convention (6–14%), not estimated line by line, and the value-added multipliers come from the OECD ICIO (2019), cross-checked against the Kazakhstan national input–output table (2023) with the same qualitative result. The FactSet and Dealroom deal extracts are pending for the revision.

# A Deal-database query specification

The same Kazakhstan deal universe is extracted from up to five commercial databases; the analysis in this draft uses the first three (Capital IQ, PitchBook, Preqin), with the FactSet and Dealroom extracts specified here and run for the final draft. Common parameters: *target/issuer geography* = Kazakhstan (country of incorporation or primary operations); *announced date* between 1 January 2015 and 31 December 2025; *deal types* = merger/acquisition, majority/minority stake, private-equity buyout and growth, venture (all stages), joint venture, and PIPE/private placement; *status* = completed or announced (pending and withdrawn recorded separately). Financials in USD at announcement.

**Capital IQ (S&P):** Screening > Transactions; Target Company Location = Kazakhstan; Transaction Types as above; Announced Date range as above. Export all columns; extraction date recorded in the replication file.

**PitchBook:** Advanced Search > Deals; Company HQ Location = Kazakhstan; Deal Types = M&A, Buyout/LBO, PE Growth/Expansion, all VC stages, Angel, Accelerator/Incubator; Deal Date range as above.

**Preqin:** Private Equity > Deals & Exits; Portfolio Company Country = Kazakhstan; Deal Date range as above.

**FactSet:** Mergers & Acquisitions / Private Markets; Target Nation = Kazakhstan; Transaction Type = as above; Announce Date range as above. *[To run.]*

**Dealroom:** Companies > HQ country = Kazakhstan; Funding rounds and M&A; Round date range as above; include grants and ICO/token as separate flags. *[To run.]*

De-duplication is on target name, announced date (±30 days) and deal type. Sector classification is harmonised to ISIC rev. 4 divisions from each database's native taxonomy via a documented crosswalk. All five extracts, the crosswalk, and the reconciliation script are in the replication package.

# References


Jean-François Arvis, Gaël Raballand, and Jean-François Marteau. *The Cost of Being Landlocked: Logistics Costs and Supply Chain Reliability*. Directions in Development — Trade. World Bank, Washington, DC, 2010.

Nicholas Bloom, Mirko Draca, and John Van Reenen. Trade induced technical change? the impact of Chinese imports on innovation, IT and productivity. *Review of Economic Studies*, 83(1):87–117, 2016.

Maxim Chupilkin, Beata Javorcik, Aleksandra Peeva, and Alexander Plekhanov. Economic sanctions and intermediated trade. *AEA Papers and Proceedings*, 115:568–572, 2025. doi: 10.1257/pandp.20251083.

Maxim Chupilkin, Beata Javorcik, and Alexander Plekhanov. The Eurasian roundabout: Trade flows into Russia through the Caucasus and Central Asia. *European Economic Review*, 187:105340, 2026. doi: 10.1016/j.euroecorev.2026.105340. Also EBRD Working Paper 276 (2023) and CEPR Discussion Paper 20097.

Gustavo de Souza, Ruben Gaetani, and Martí Mestieri. More trade, less diffusion: Technology transfers and the dynamic effects of import liberalization. Working Paper 2024-20, Federal Reserve Bank of Chicago, 2024. Revised 2026; SSRN 4972150.

Avinash K. Dixit and Robert S. Pindyck. *Investment under Uncertainty*. Princeton University Press, Princeton, NJ, 1994.

Raymond Fisman and Shang-Jin Wei. Tax rates and tax evasion: Evidence from "missing imports" in China. *Journal of Political Economy*, 112(2):471–496, 2004. doi: 10.1086/381476.

Gita Gopinath and Brent Neiman. Trade adjustment and productivity in large crises. *American Economic Review*, 104(3):793–831, 2014. doi: 10.1257/aer.104.3.793.

Robert C. Johnson and Guillermo Noguera. Accounting for intermediates: Production sharing and trade in value added. *Journal of International Economics*, 86(2):224–236, 2012.

Réka Juhász. Temporary protection and technology adoption: Evidence from the Napoleonic blockade. *American Economic Review*, 108(11):3339–3376, 2018. doi: 10.1257/aer.20151730.

Réka Juhász, Nathan Lane, and Dani Rodrik. The new economics of industrial policy. *Annual Review of Economics*, 16:213–242, 2024. doi: 10.1146/annurev-economics-081023-024638. NBER Working Paper 31538 (2023).

Tarun Khanna and Krishna Palepu. Why focused strategies may be wrong for emerging markets. *Harvard Business Review*, 75(4):41–51, 1997.

Robert Koopman, Zhi Wang, and Shang-Jin Wei. Tracing value-added and double counting in gross exports. *American Economic Review*, 104(2):459–494, 2014. doi: 10.1257/aer.104.2.459.

Raghuram G. Rajan and Luigi Zingales. Financial dependence and growth. *American Economic Review*, 88(3):559–586, 1998.